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| United States Patent | 12404600            |
| Kind Code            | B2                  |
| Date of Patent       | September 02, 2025  |
| Inventor(s)          | Cohen; Peter et al. |

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### Fins on crucible or gussets on refractory lining for facilitating exclusion of impurities from a boule

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#### Abstract

A furnace and a method of growing a high temperature oxide crystal in the furnace. The furnace includes a crucible having a melt therein and a heating element for generating heat in the melt. A thermal element within the furnace produces a thermal gradient within the melt to draw a cold spot of a convection cell of the melt away from a seed location of the crucible. A seed crystal is drawn from the melt at the seed location to form a boule to grow the high temperature oxide crystal.

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**Appl. No.:** 18/326342

**Filed:** May 31, 2023

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20240401227 A1   | Dec. 05, 2024    |

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#### Publication Classification

**Int. Cl.:** C30B15/10 (20060101); C30B15/14 (20060101); C30B29/16 (20060101)

**U.S. Cl.:**

CPC      **C30B15/10** (20130101); **C30B15/14** (20130101); **C30B29/16** (20130101);  
Y10T117/1016 (20150115); Y10T117/1032 (20150115)

**Field of Classification Search**

**CPC:**    C30B (15/10); C30B (15/14); C30B (29/16)

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**Background/Summary**

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates generally to crystal growth and in particular relates to methods for reducing impurities in high temperature oxide crystals grown for scintillation detectors used in Positron Emission Tomography (PET).

Description of the Related Art

(2) In manufacturing a boule of Lutetium Oxyorthosilicate (LSO), Lutetium Yttrium Oxyorthosilicate (LYSO) or Gadolinium Aluminium Gallium Garnets (GAGG) type scintillators, powders are added to a crucible in an appropriate stoichiometric ratio and heated to form a melt. A seed crystal is then lowered to contact the melt and gradually drawn away from the melt to form the boule. It is desirable for any dopants in the melt to be evenly distributed as the boule is drawn so that the distribution of the dopants in the boule is uniform. It is also desirable for impurities to not be incorporated in the boule.

(3) However, this is typically not the case. Due to the geometrical symmetry of the furnace and the

thermal field it produces, a symmetrical flow field of the melt produces a cold spot in the center of the crucible, creating a location at which impurities can collect. Since the seed crystal contacts the melt in the center (i.e., at the cold spot), the impurities tend to become incorporated into the boule. It is therefore desirable to create a flow field in the crucible that moves impurities away from a location at which a seed crystal contacts the melt.

#### BRIEF SUMMARY OF THE INVENTION

(4) Disclosed herein is a method of growing a high temperature oxide crystal; wherein the method includes disposing a melt within a crucible, the crucible including at least one thermal element, heating the melt within the crucible, wherein the at least one thermal extension produces a thermal gradient within the melt to draw a cold spot of a convection cell of the melt away from a seed location of the crucible, and drawing a boule from the seed location of the crucible to grow the high temperature oxide crystal.

(5) Disclosed herein also is a furnace for growing a high temperature oxide crystal; the furnace including a crucible having a melt therein, a heating element for generating heat in the melt, a thermal element within the furnace that produce a thermal gradient within the melt to draw a cold spot of a convection cell of the melt away from a seed location of the crucible, and a seed crystal for drawing a boule from the melt at the seed location to grow the high temperature oxide crystal.

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## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects, and advantages of the present invention will become better understood with reference to the following description and appended claims, and accompanying drawings where:

(2) FIG. 1 is a side view of a furnace in which a boule is grown, in an illustrative embodiment;

(3) FIG. 2 shows a side view of a crucible in the furnace during a heating of the melt;

(4) FIG. 3 shows a top view of the furnace depicting thermal elements used to produce a thermal gradient in the crucible;

(5) FIG. 4 shows a side view of the crucible, in another embodiment;

(6) FIG. 5 shows a top view of a crucible shown in FIG. 4;

(7) FIG. 6 shows a side view of the crucible of FIG. 5 during a formation of a boule; and

(8) FIG. 7 shows a side view of the crucible 114 with a lid.

(9) It should be understood that the various embodiments are not limited to the arrangements and instrumentality shown in the drawings.

#### DETAILED DESCRIPTION OF THE INVENTION

(10) The present invention may be understood more readily by reference to the following detailed description of preferred embodiments of the invention as well as to the examples included therein. All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure.

(11) FIG. 1 is a side view of a furnace 100 in which a boule is grown, in an illustrative embodiment. The furnace 100 includes a furnace wall 102 enclosing a space 104. A copper coil 106 is wrapped around an outer surface of the furnace wall 102. Water or coolant flows through the copper coil 106 to carry heat away from the furnace 100 or maintain a temperature in the space 104 within the furnace 100. A quartz tube 108 is disposed concentrically within the furnace wall 102. An induction coil 110 disposed is in an annulus between the quartz tube 108 and the furnace wall 102. A crucible 114 is placed at a bottom surface 116 of the furnace 100 within the quartz tube 108. The crucible 114 includes a right cylindrical wall extending upward from a bottom crucible surface.

The crucible **114** can be made of iridium. The crucible **114** extends partway up the furnace **100** from the bottom surface **116**. A brick cylinder or ceramic cylinder **115** extends from a top of the crucible **114** to a top or lid **130** of the furnace **100**. A thermal filler **118** is disposed in an annulus **128** between the quartz tube **108** and the crucible **114**/brick cylinder **115**.

(12) To make a boule, ingredients for forming a melt **120** are disposed in the crucible **114** and the crucible **114** is placed concentrically within the furnace **100**. The induction coil **110** is activated to heat the ingredients to form the melt **120**. A seed crystal **122** is lowered through the brick cylinder **115** at an end of a rod **124** to contact a top of the melt **120** at a seed location (i.e., a location at which the boule is to be drawn from the melt). The seed crystal **122** is then drawn upward (with accompanying rotary motion) from the melt **120**, allowing the melt **120** to cool and crystallize around the seed crystal **122**. Drawing the seed crystal **122** upward therefore creates a boule or cylindrical crystal.

(13) FIG. 2 shows a side view **200** of the crucible during a heating of the melt. Convection cells **202** form in the melt **120** when the melt heated above a given temperature. The heat at the wall **204** of the crucible **114** causes the melt **120** to flow upward at the wall **204**. Upon reaching the top of the crucible **114**, the melt **120** migrates radially inward toward a central axis **206** of the crucible **114**, cooling in the process. A cold spot **215** thus develops at the top of the melt at the central axis **206**. Impurities will tend to collect at the cold spot **215**. The melt **120** then descends to the bottom of the crucible **114** along the central axis **206**. The method disclosed herein creates a thermal gradient in the melt to move the cold spot **215** away from the central axis **206**.

(14) FIG. 3 shows a top view **300** of the furnace depicting thermal elements used to produce a thermal gradient in the crucible **114** and thus throughout the melt **120** within the crucible **114**. The crucible **114** includes at least one thermal element, such as a fin **302**, on an its outer surface. For illustrative purposes, four fins **302a-d** are shown on the outer surface of the crucible **114**, separated from each other by 90 degrees. However, in various embodiments, any number of fins, including only one fin, can be formed on the crucible **114** and the spacing can be fins can be any amount and the fins need not be equally spaced. Referring to fin **302a** for illustrative purposes, fin **302a** is in contact with the crucible **114** at a selected location of the outer surface of the crucible **114**. The fin **302a** radiates heat away from the crucible and thus lowers a local temperature of the melt **120** at a location **310** inside the crucible **114** that is proximate the location at which the fin **302a** is in contact with the crucible **114**. The amount by which the local temperature is lowered is related to a thermal resistance of the fin which is based on a size of the fin, a material of the fin, a shape of the fin, etc. As shown in FIG. 3, the fins **302a-d** can have different thermal resistances, i.e., different sizes, material, shapes, etc. For example, the first fin **302a** can have a first thermal resistance and the second fin **302b** can have a second thermal resistance different from the first thermal resistance. The design and placement of the fins **302a-d** can be selected to create a thermal gradient that causes the cold spot to migrate to a selected location away from the central axis **206** of the crucible **114**.

(15) In another embodiment, the thermal filler **118** between the quartz tube **108** and the crucible **114** can include a thermal element, such as one or more ceramic gussets **304a-d** extending inward from an outer surface **306**. The size, shape and material of the one or more gussets **304a-d** affects local temperatures within the melt and therefore the formation of a thermal gradient within the melt. For example, the first gusset **304a** can have a first thermal resistance and the second gusset **304b** can have a second thermal resistance different from the first thermal resistance. The gussets **304a-d** radiate heat asymmetrically into the crucible, thereby affect the location of the cold spot **215**. In various embodiments, both fins **302a-d** and gussets **304a-d** can be used together to create the thermal gradient within the melt **120**.

(16) FIG. 4 shows a side view **400** of the crucible **114**, in another embodiment. The crucible **114** includes a thermal element such as one or more baffles **402a**, **402b** extending radially inward from the walls of the crucible **114**. The baffles **402a**, **402b** redirect a flow of the melt **120** within the

crucible **114** away from the standard convection cell shown in FIG. 2. The transverse cross-section of the first baffle **402a** is thicker than transverse the cross-section of the second baffle **402b**. As a result of this difference, the first baffle causes greater heating than the second baffle, causing a thermal gradient in the direction between the first baffle and the second baffle, thereby facilitating the formation of an asymmetric convective flow that moves the location of the cold spot **125** away from a central axis **206** of the crucible **114**.

(17) FIG. 5 shows a top view **500** of the crucible **114** shown in FIG. 4. The baffles **402a**, **402b** extends vertically parallel to the central axis **206** along the inner wall of the crucible **114**. Each baffle **402a**, **402b** projects radially inward from the inner surface of the wall. The amount by which the baffles project inward varies with depth. The amount of inward projection is greatest at the bottom of the crucible **114** and decreases with height in the crucible **114** until they are flush or almost flush with the inner wall at the top of crucible **114**. As a result of this configuration, the baffles **402a**, **402b** are more intrusive in the convection flow cells at the bottom of the crucible **114**, thereby creating more stirring to occur at the bottom of the melt **120** than at the top of the melt **120**. This stirring, as well as the asymmetric heating, affects the shape of the convection cell and thus the location of the cold spot **215** as seen in the FIG. 4.

(18) FIG. 6 shows a side view **600** of the crucible of FIG. 5 during a formation of a boule **602**. Due to the configuration of the baffles **402a**, **420b**, the cold spot **215** is to one side of the central axis, as shown in FIG. 6. The rod **124** (and seed crystal) can be lowered into the crucible **114** at a location that is eccentric from the central axis **206** of the crucible **114** and away from the cold spot **215**. In another embodiment, a central axis of the rod **124** (and seed crystal) can be aligned with the central axis of the crucible **114**. The eccentric location of the cold spot **215** (away from the center) will ensure

(19) FIG. 7 shows a side view **700** of the crucible **114** with the lid **130**. The lid **130** includes a thermal element such as baffles **702a**, **702b** extending downward from a bottom surface of the lid **130** and into the crucible **114** and the melt **120**. The baffles **702a**, **702b** are diametrically opposite from each other and have a radially inward projection extending inward from an outer edge of the lid **130**. Similar to the baffles **402a**, **402b** of FIGS. 4 and 5, the size, shape and material of the baffles **702a**, **702b** can be different. The radially inward projection of the baffles **702a**, **720b** can create a thermal gradient in the melt **120**, thereby causing the cold spot **215** of the melt **120** to move away from a central axis **206** of the crucible **114**.

(20) Although the present invention has been described in considerable detail with reference to certain preferred versions thereof, other versions are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the preferred versions contained herein.

(21) The reader's attention is directed to all papers and documents which are filed concurrently with this specification, and which are open to public inspection with this specification, and the contents of all such papers and documents are incorporated herein by reference.

(22) All the features disclosed in this specification (including any accompanying claims, abstract, and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

(23) Any element in a claim that does not explicitly state "means for" performing a specified function, or "step for" performing a specific function, is not to be interpreted as a "means" or "step" clause as specified in 35 U.S.C § 112, sixth paragraph. In particular, the use of "step of" in the claims herein is not intended to invoke the provisions of 35 U.S.C § 112, sixth paragraph.

## Claims

1. A furnace for growing a crystal, comprising: a crucible having a melt therein, the crucible having right cylindrical inner surface; a first baffle projecting radially inward from the right cylindrical

inner surface at a first location and a second baffle projecting radially inward from the inner surface at a second location diametrically opposite the first baffle, wherein an amount by which the first baffle and the second baffle project radially inward from the right cylindrical inner surface is greatest at a bottom of the crucible and decreases with height to be flush with the right cylindrical inner surface at a top of the crucible, wherein a first transverse cross-section of the first baffle is different from a second transverse cross-section of the second baffle; a heating element for generating heat in the melt; and a rod for drawing the crystal from the melt at a seed location to grow the crystal.

2. The furnace of claim 1, further comprising a thermal element, wherein the thermal element includes at least one fin extending from an outer surface of the crucible.

3. The furnace of claim 2, wherein the at least one fin includes a first fin having a first thermal resistance at a first location of the outer surface and a second fin having a second thermal resistance at a second location of the outer surface.

4. The furnace of claim 1, further comprising a thermal element, wherein the thermal element comprises at least one gusset extending inward from a thermal filler surrounding the crucible.

5. The furnace of claim 4, wherein the at least one gusset includes a first gusset having a first thermal resistance at a first location of the thermal filler and a second gusset having a second thermal resistance at a second location of the thermal filler.

6. The furnace of claim 1, wherein the first baffle causes greater heating in the melt than the second baffle.

7. The furnace of claim 1, further comprising a thermal element, wherein the thermal element comprises a lid to the crucible having a first baffle and a second baffle extending from the lid into the melt.

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